

Human-Computer Interaction

Johan F. HOORN

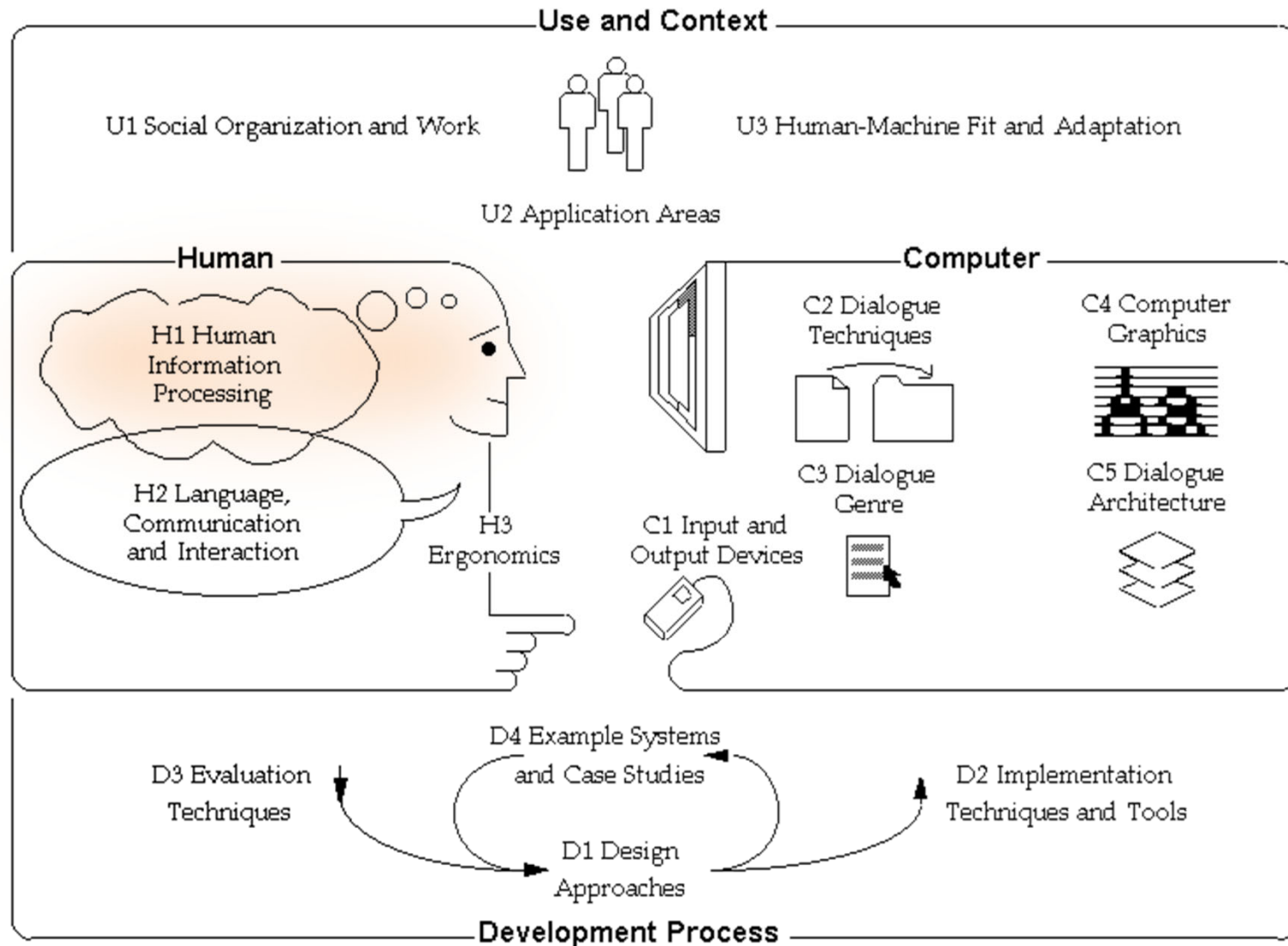
Lecture 02: Human Information Processing, Part 2



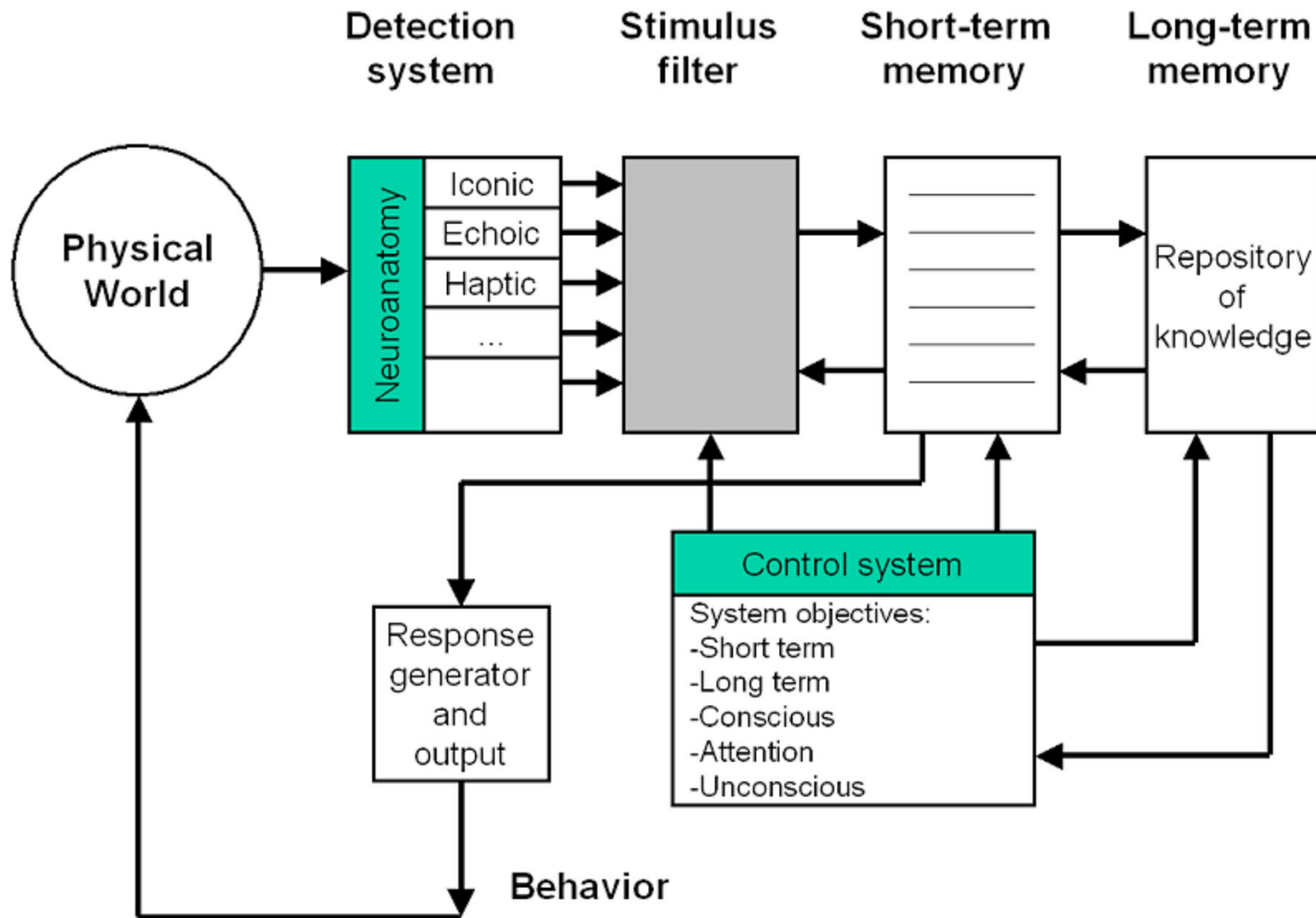
C-3PO ("3PO" aka Threepio)

Lukas, G., & Hales, J. (2002). *Star Wars Episode II: Attack of the Clones*. Los Angeles, CA: 20th Century Fox. [Part 7/10] - 1080p HD, 7:00 min. Retrieved May 21, 2012 from http://www.youtube.com/watch?v=-o5Y_AszGg

Overview



The Human Information Processor



Problem representation

How you represent a problem (e.g., text or image) influences its solution

Text works better
to explain
abstractions:
'Infinity'
'Intelligence'

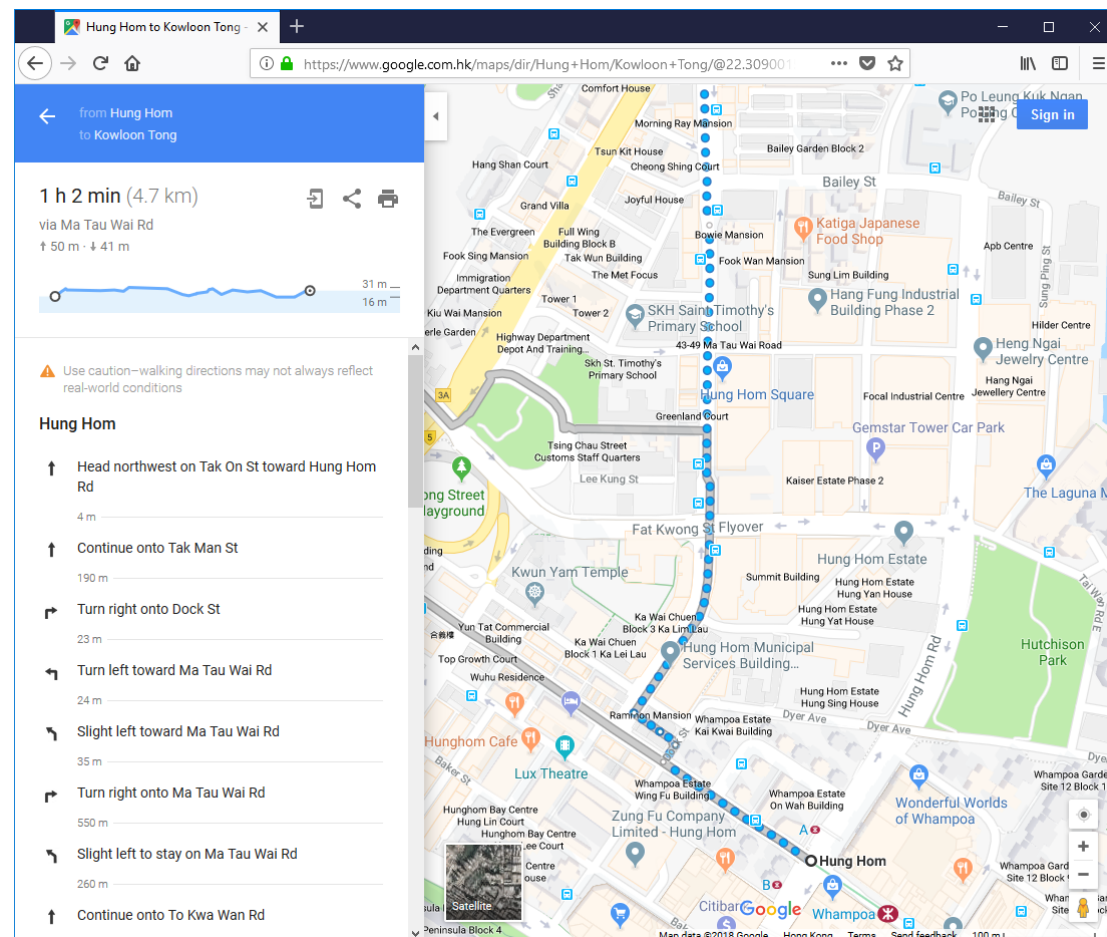


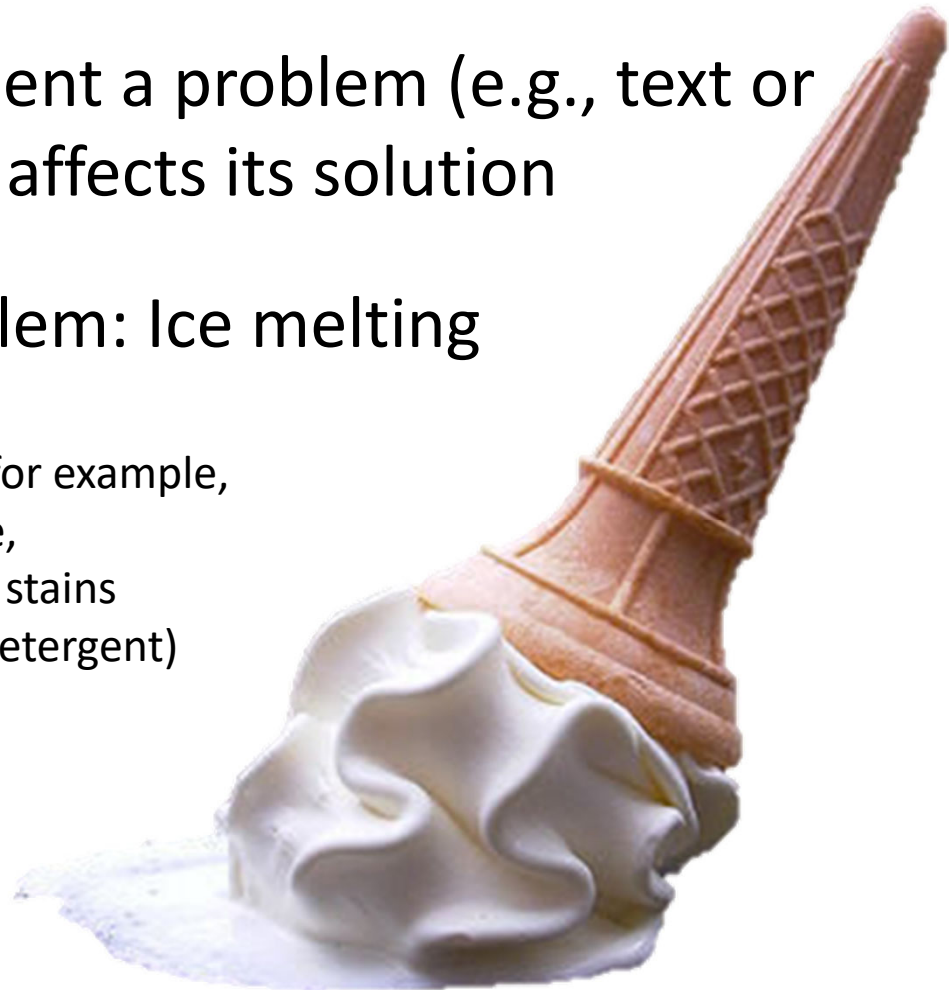
Image works
better for spatial
orientation and
localisation. Where
is what?

Problem representation

How you represent a problem (e.g., text or image) affects its solution

Problem: Ice melting

Solutions become focused on, for example,
colour changes, shapes, texture,
deformation of solids, sun, and stains
(walk in the shadow, napkins, detergent)



Surface level
Phenomenology

Problem representation

How you represent a problem (e.g., text or image) affects its solution

Problem: Ice melting



Solutions directed to fixing loss of structure without creating gritty ice crystals and managing processes and mechanisms of, for instance, atoms and ions (e.g., add emulsifiers, more sodium chloride)

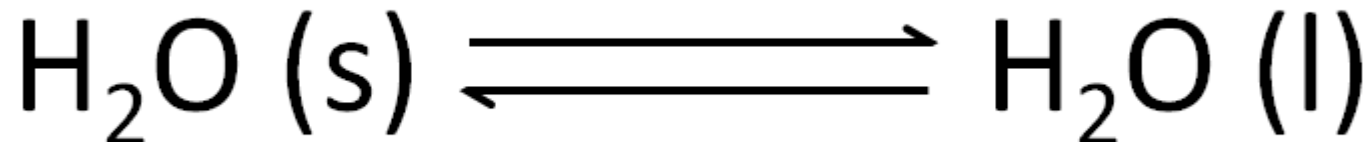
Structural level

Problem representation

3

How you represent a problem (e.g., text or image) affects its solution

Problem: Ice melting



Solutions derived from logics, reasoning, abstraction.

Protein BsIA binds with N_2 and O_2 (air), H_2O , and $-\text{COOH}$ (component of fat)

Symbolic level

7

Global warming solved: Scientists find ingredient that stops ice cream melting





VIDEO GAMES

MOVIES & TV

COMICS & COSPLAY

TABLETOP

SCIENCE & TECH

THE ESCAPIST PORTAL > THE ESCAPIST FORUMS > THE NEWS ROOM

Global Warming Solved: Scientists Find Ingredient That Stops Ice Cream Melting

THREAD INDEXSEARCH FORUMSCONTACT MODERATORSPOSTING GUIDELINES

PatrickJS

Muckraker





Posts: 260

Joined: 8 Jun 2015

1 POSTED: 1 September 2015 6:43 am

Global Warming Solved: Scientists Find Ingredient That Stops Ice Cream Melting

British scientists have isolated an ingredient that can keep ice cream from melting in the summer heat.

As the [U.N. convenes a closed-door meeting on climate change](#) this week, British scientists are one step ahead of the game on averting the worst kind of heat-related catastrophe: next summer, [your ice cream cone won't melt](#).

[Food researchers](#), from the Universities of Dundee and Edinburgh, have isolated a naturally occurring protein, BsIA, that binds the air, fat and water in your triple scoop. In the end, this severely limits how quickly your frozen treat is going to melt.

The protein also keeps the ice cream frozen for longer, preventing gritty ice crystals from forming when you've kept the stuff in your freezer too long.

If you are interested in more nutritious food science news, you might like to [sample some bacon-flavored seaweed](#).

Cait MacPhee, a professor of [ice cream science](#) at Edinburgh and leader of the study, told the Telegraph, "we're excited by the potential this new ingredient has for improving ice cream, both for consumers and for manufacturers."

Personally, I'm not sure what the big deal is here. People have told me of this supposed 'melting,' but I've eaten three bowls of ice cream since I started writing this article and haven't noticed a thing.

Now, if only someone could invent ice cream that doesn't fall out of the cone just after you've paid for it...

Source: [Telegraph.co.uk](#)

[Permalink](#)



Image Source: McGill University

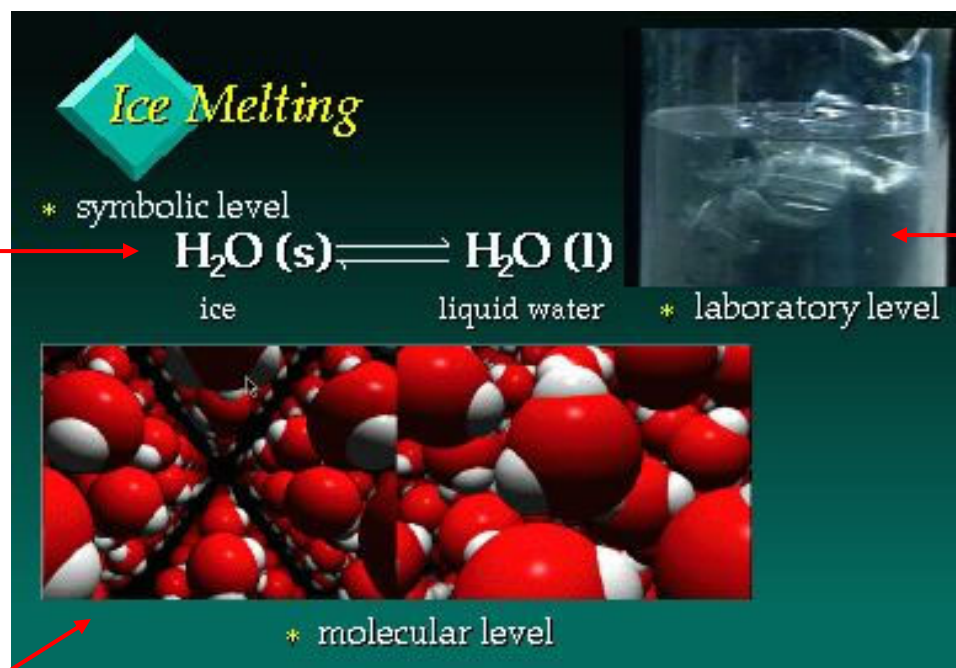
Solkard

2 POSTED: 1 September 2015 7:10 am

Problem representation

How you represent a problem affects its solution,
ALSO IN USER INTERFACES!

Symbolic level
emphasises
form transition from
solid to liquid matter
as logic reasoning



Visible laboratory level
emphasises e.g., colour
changes, formation of
solids, and boiling

Molecular level
(Quicktime animation)
emphasises changes in structure and processes of,
for instance, atoms and ions

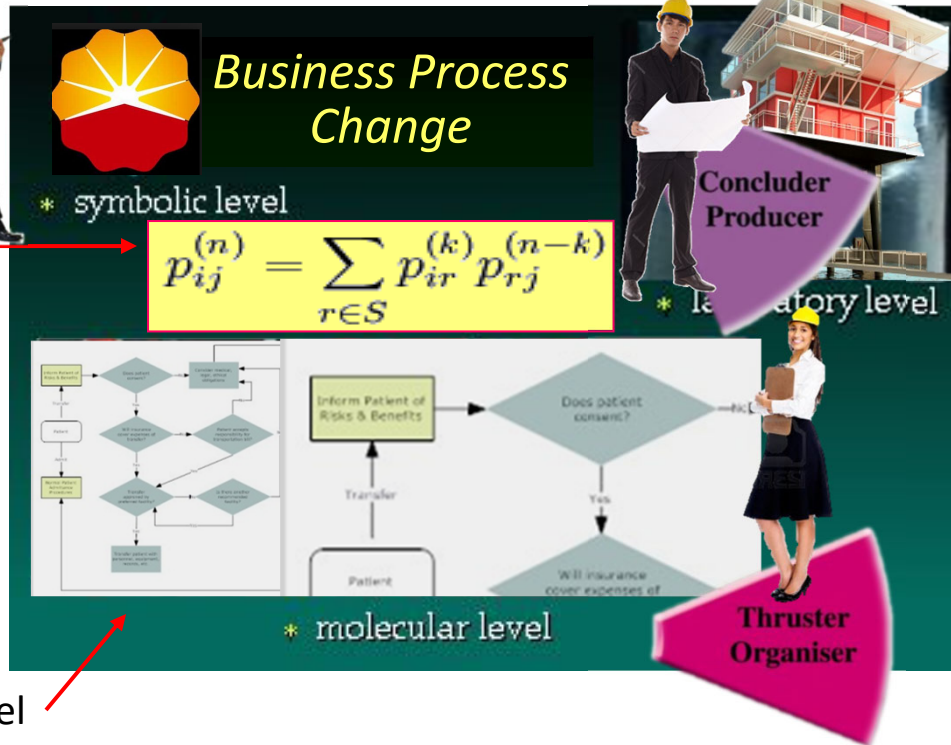
VisChem <http://www.uws.edu.au/science/Staff/rtasker/papers/images/interface.gif>

Change one level
and see how the
other levels change
as well. Insight!

Make multiple representations, each addressing another aspect

Problem: I want to solve this optimisation problem

Symbolic level
emphasises
transition from
input to output
as logic reasoning



Visible laboratory level
emphasises,
for example,
type of workforce,
unforeseen mishaps,
work culture

Structural level
(Quicktime animation)

emphasises
changes in structure and processes of,
e.g., work units and their actions

So to improve the production, three sub-goals
took three different representations

As a designer, your clients will represent their problems in one specific way
Draw in more information than the problem representation that your client offers!

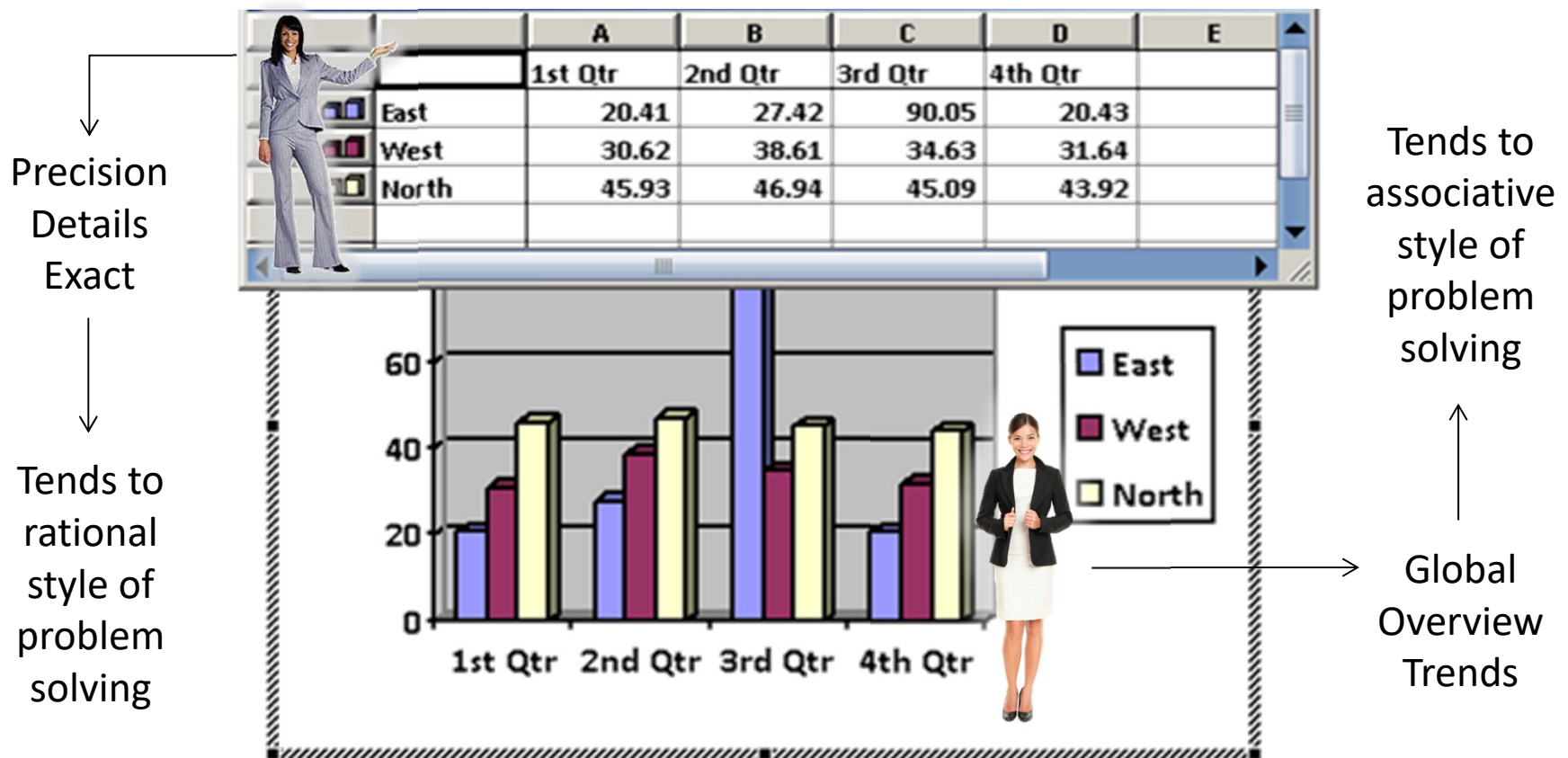
You copied everything from the Internet
We have a legal issue here

Let's try a new business model



In sum, how you represent a problem to your client (or your client to you), begs its solution

Each level of aggregating the data (cf. goal-decomposition) requires its own form of representation (cf. design)



Microsoft Word chart editor

Different mind sets solve problems differently

Who is your client (and who are you)?

The logic rationalist

Feeling happy in closed systems,
like a fish in a tank



The associative empiricist

Feeling happy in open systems,
like a bird in a tree

Our students

The associative empiricist

Feeling happy in open systems

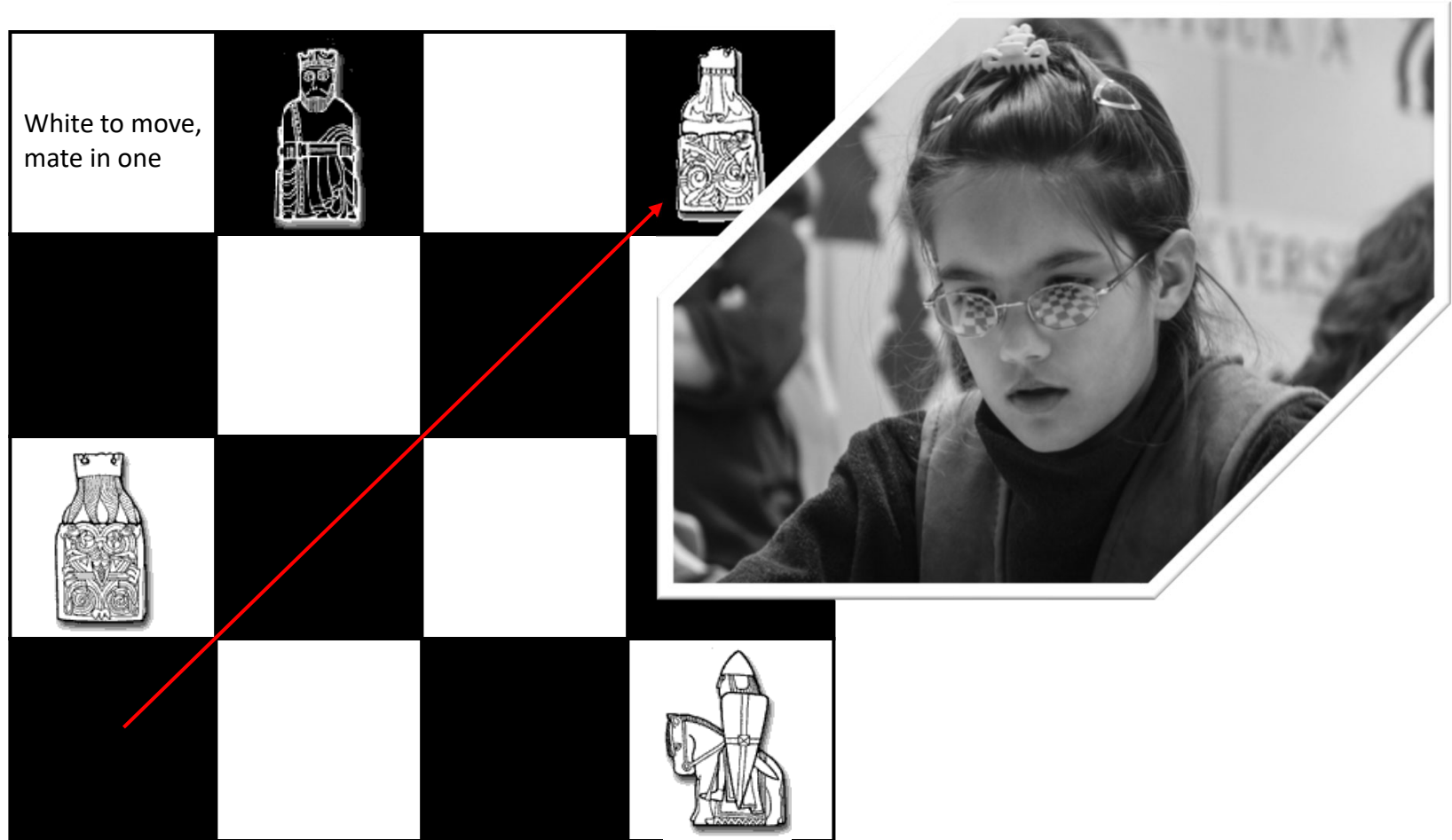


The logic rationalist

Feeling happy in closed systems



Structured decisions usually relate to well-defined problems and a more precise problem state



<http://www.w-isles.gov.uk/w-isles/museum/gallery.htm>

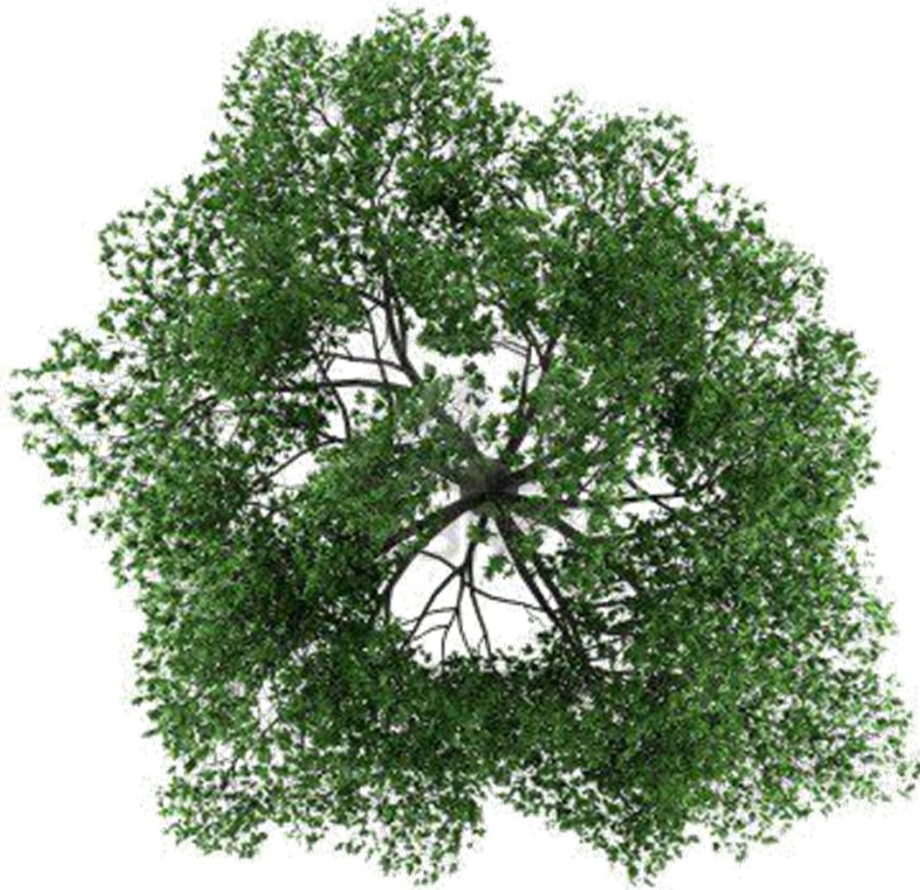
Unstructured decisions usually relate to ill-defined problems and a more vague problem state



What are you drawing?

Don't know.
I'm drawing.

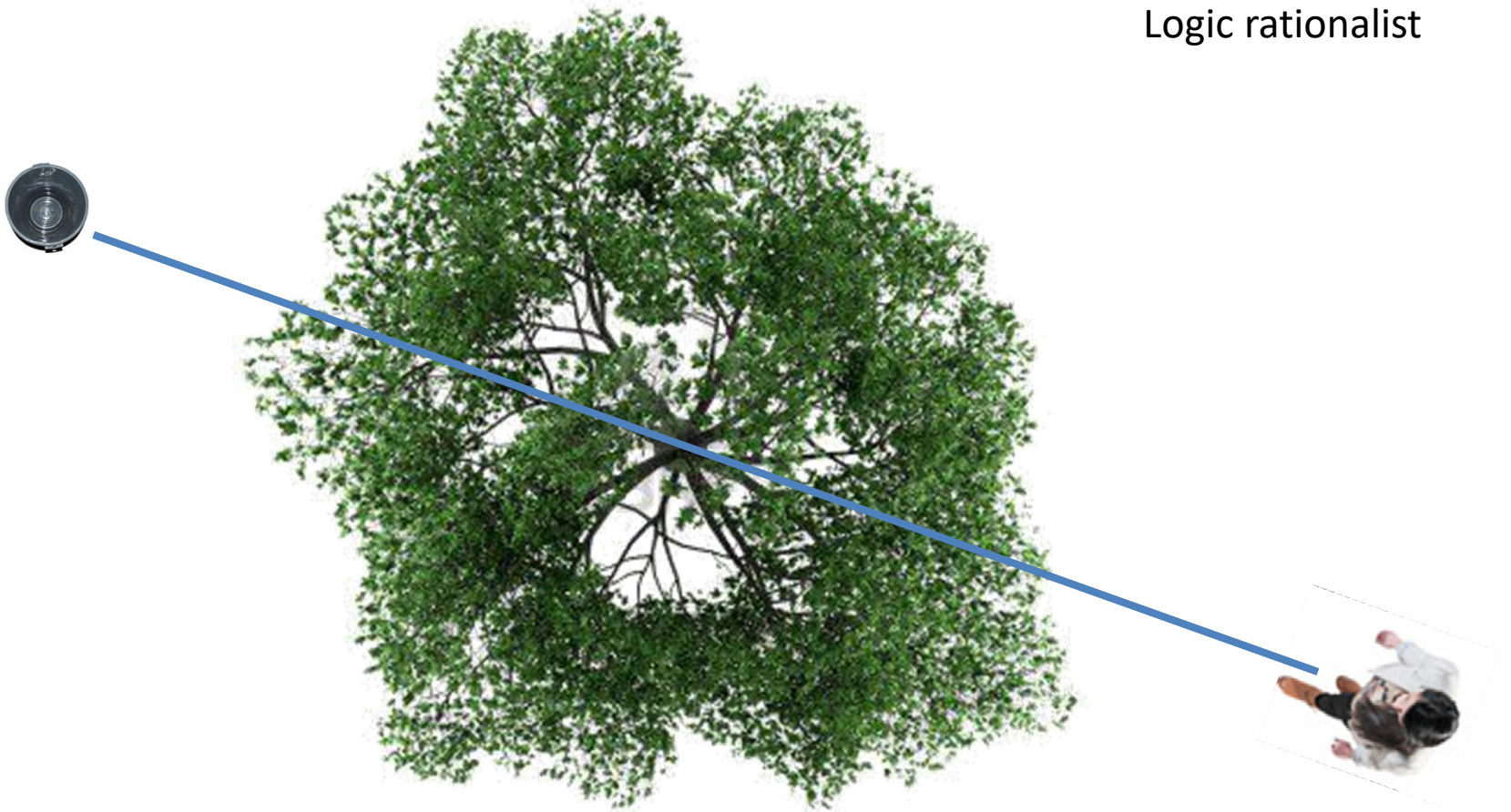
Tree - bin - me



Position yourself such that the bin, the tree, and you form a straight line

Tree - bin - me

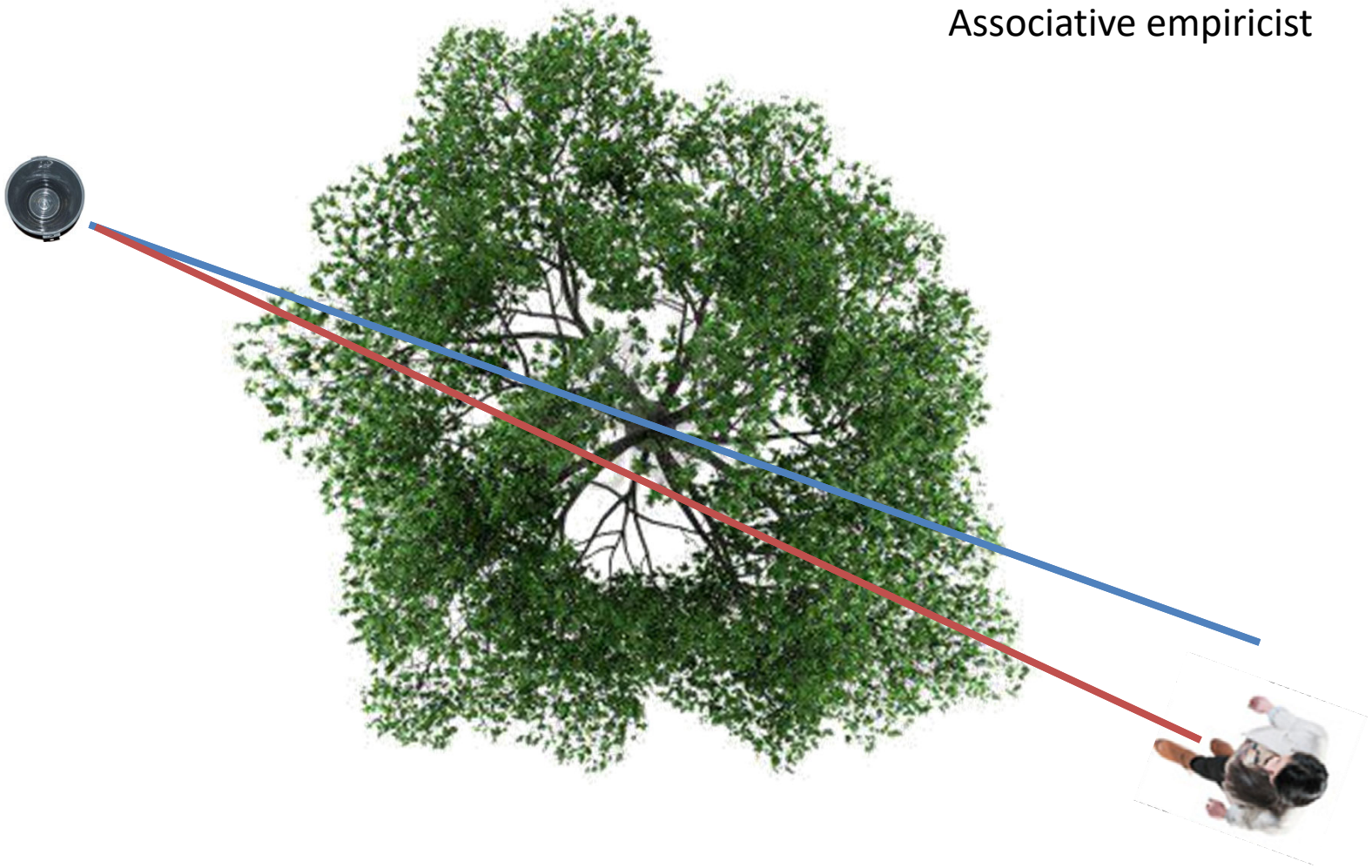
Logic rationalist



Position yourself such that the bin, the tree, and you form a straight line

Tree - bin - me

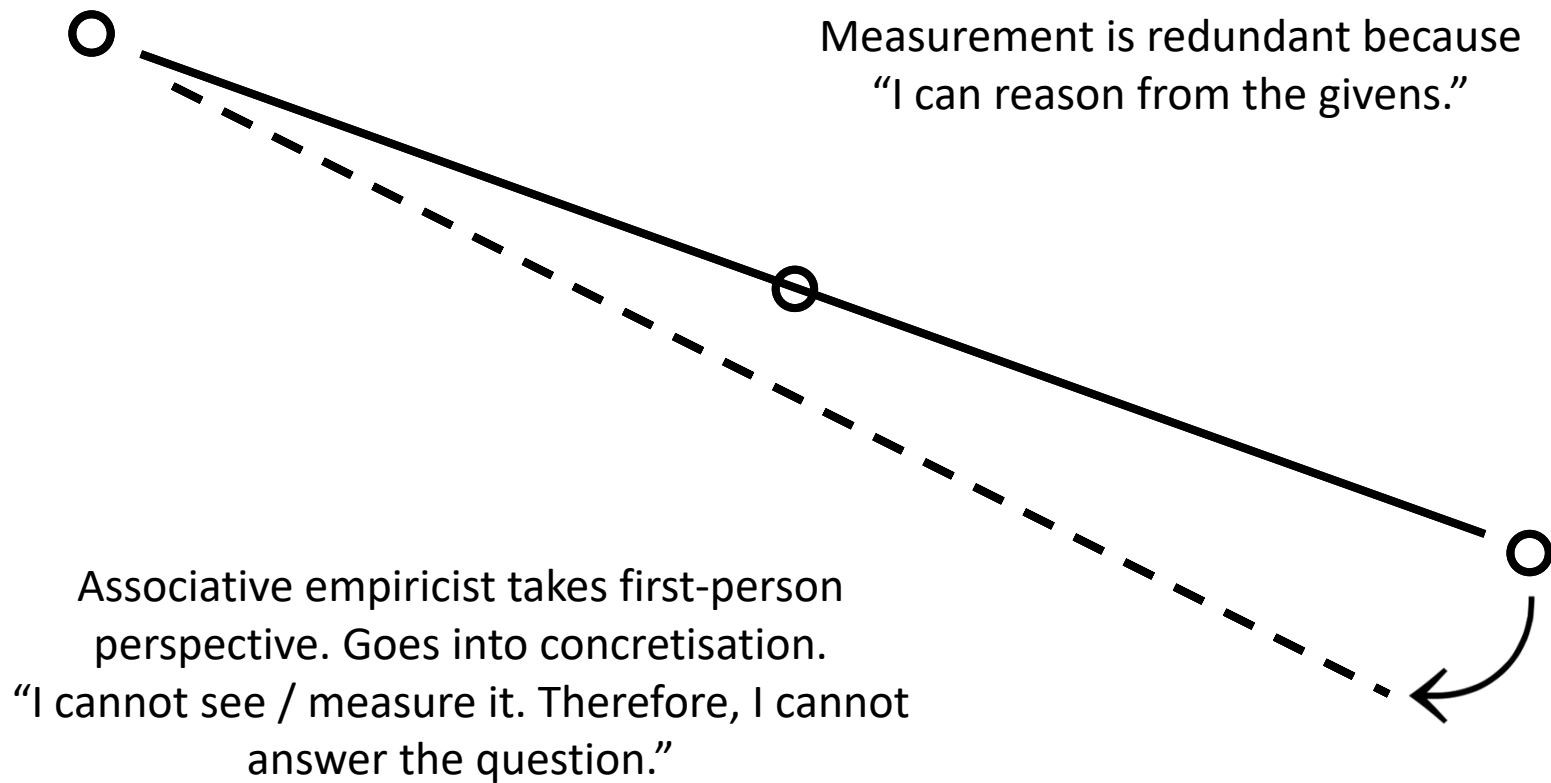
Associative empiricist



Position yourself such that the bin, the tree, and you form a straight line

Tree - bin - me

Logic rationalist takes person-independent overview.
Sees him/herself as a point in space.
Abstracts from the real world.
Measurement is redundant because
“I can reason from the givens.”



By now you probably think that
associative empiricists
are pretty stupid



Let's see:

How many blocks in this construction?

1st layer $1+3+4+5+5$

2nd layer $1+2+3+4$

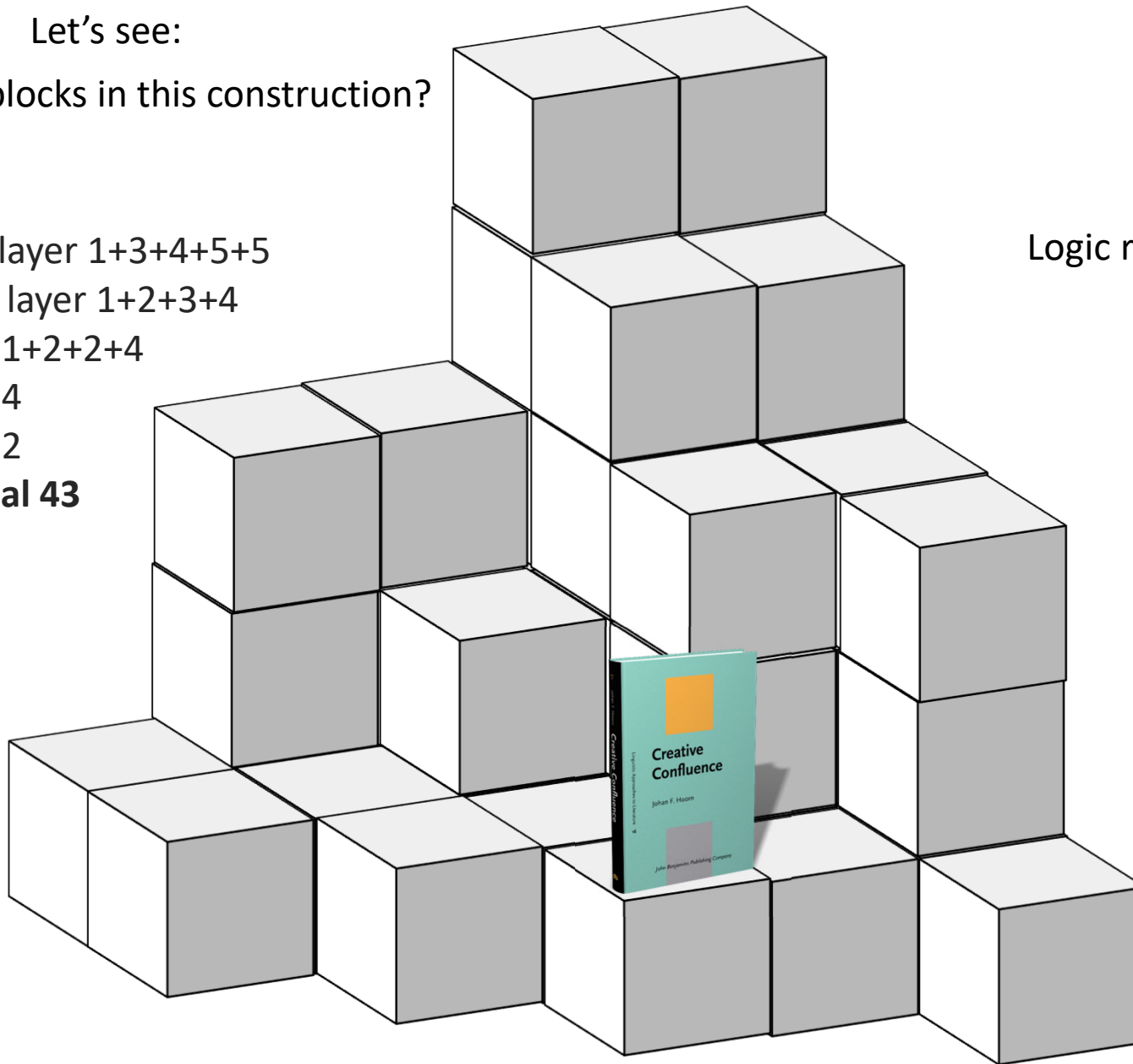
3rd $1+2+2+4$

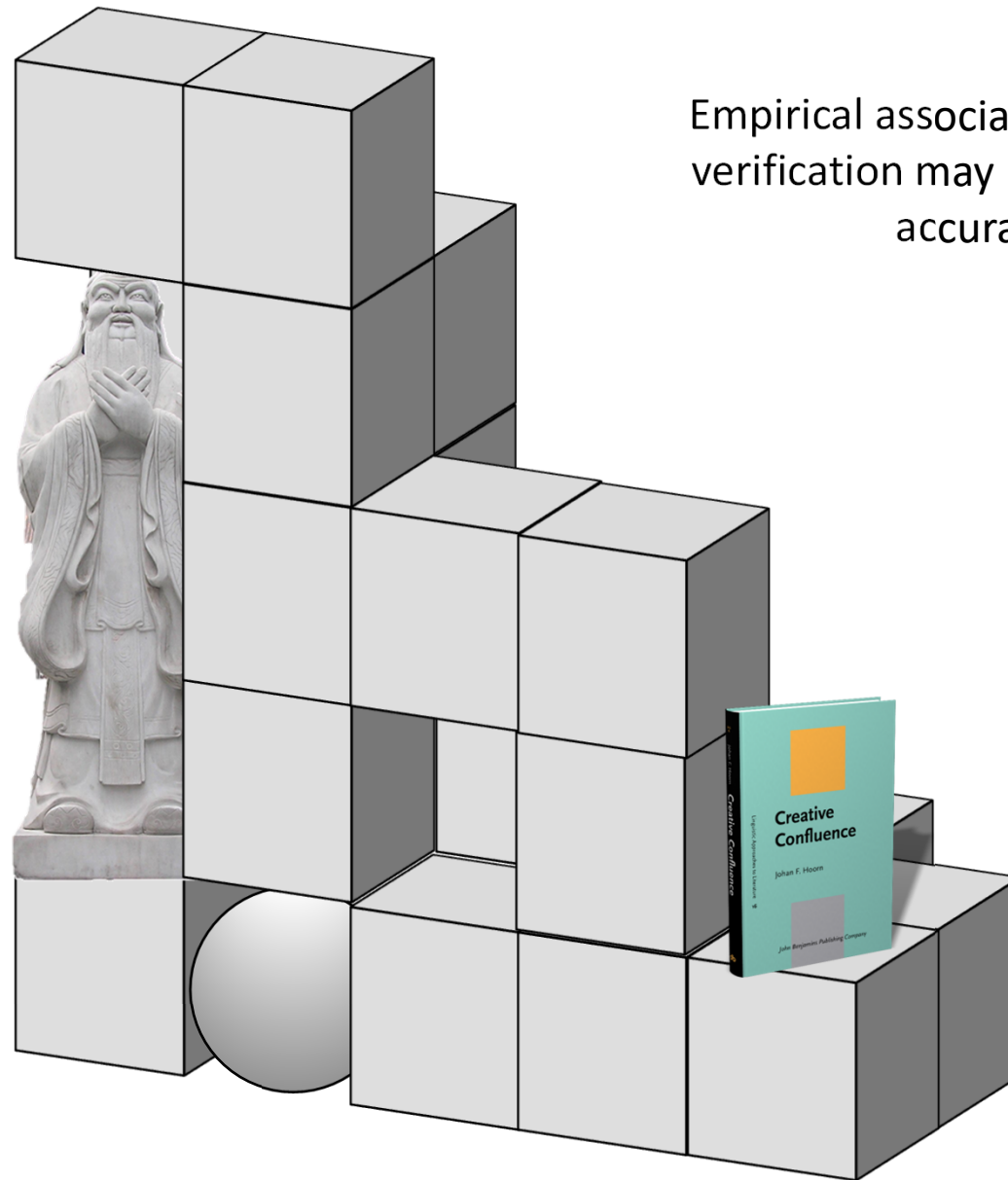
4th 4

5th 2

Total 43

Logic rationalism

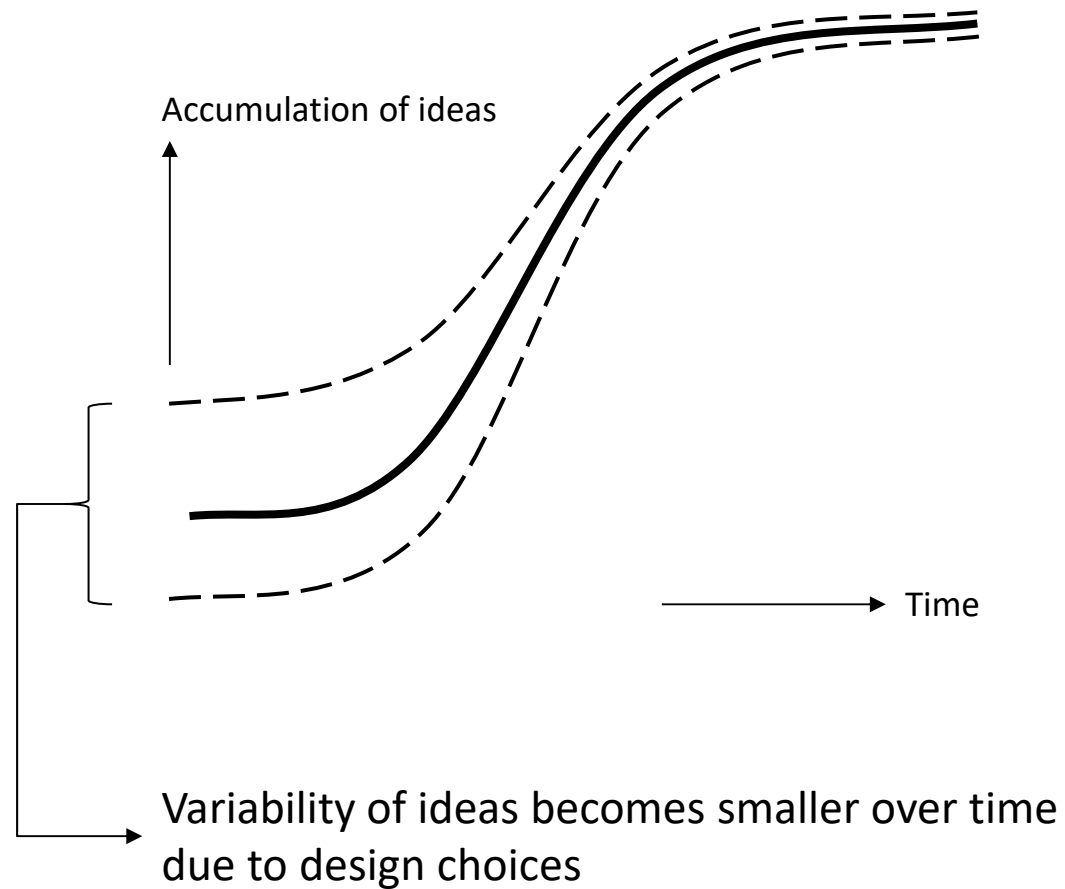




Empirical association and then
verification may lead to better
accuracy

Our students working together

Product life-cycle follows the Innovation S-curve



Break (5 minutes)



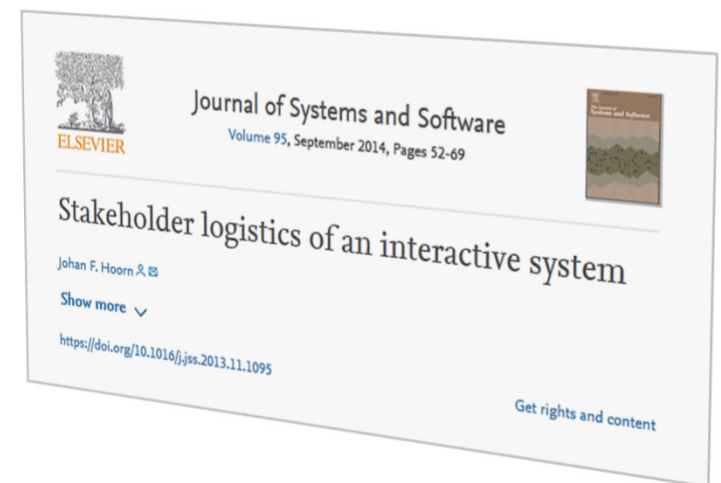
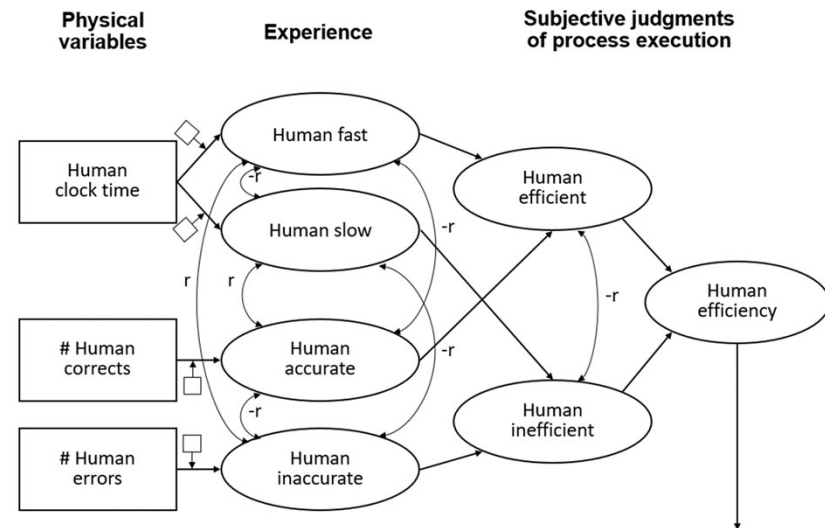
Output: response

❑ Motor

- Speed-accuracy trade-off
 - Fitts' law
 - Shannon's theory
- Performance evaluation
 - Index of Difficulty
 - Index of Performance

Reaction times

- Measuring the speed of cognitive processing
- Assessing the complexity of mental operations
- Investigating the effects of practice and learning on task performance
- Exploring decision-making processes
- Evaluating attentional mechanisms and response selection
- Comparing cognitive performance across different populations (e.g., age groups, clinical vs. non-clinical)



Output: response

☐ Motor

Fitts' Law

Prof. Paul Morris Fitts was Lieutenant-colonel of the US Air Force, Head of US Air Force Human Engineering Division, later professor at Ohio State U and U of Michigan

Fitts' law (1954) states that the amount of time required for a person to move a pointer to a target area (e.g., finger to button) is a function of the distance to the target divided by the size of the target. Thus, the longer the distance and the smaller the target's size, the longer it takes

Thus, Fitts' Law relates to the Efficiency of system performance



Cockpit controls of a Spitfire fighter

Output: response

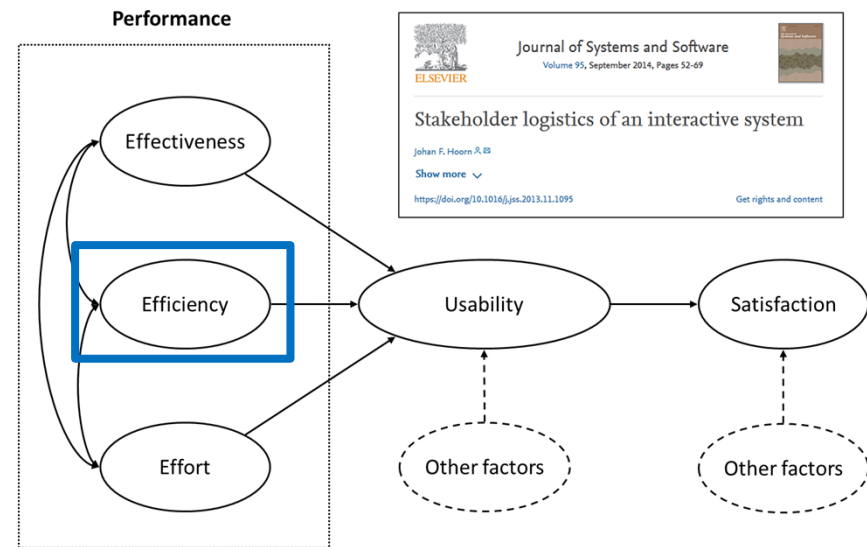
❑ Motor

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Fitts measures physical side or experience side of performance?

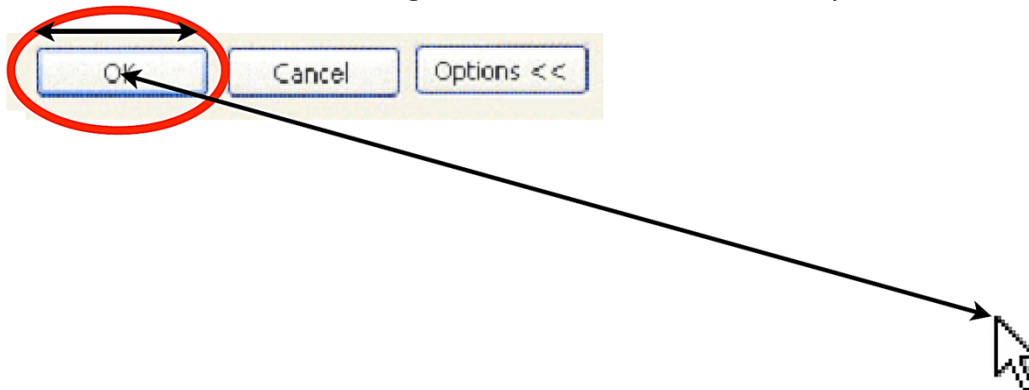
Output: response

❑ Motor

Fitts' Law

- A model of human movement
- Predicts the time required to rapidly move from a starting position to a final target area
- Models the act of pointing
- In the real world (e.g., with a hand or finger, or even eye tracking)
- On computers (e.g., with a mouse or a joystick)

The wider an interface widget and the closer it is to the pointer, the easier to hit it



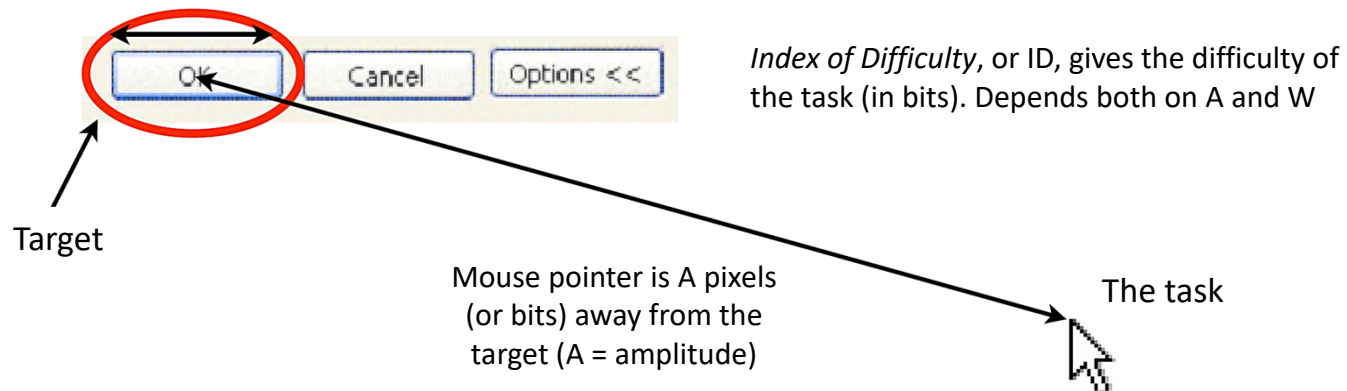
Output: response

❑ Motor

Fitts' Law

- Based on Shannon's theory of information systems
- Human motor system modelled as channel used to transmit information
- Task has a certain difficulty (measured by number of bits)
- Channel has a certain bandwidth (measured by seconds/bit)

Target is W pixels wide (W = tolerance)



The logarithm in Fitts' Law is there because of how human movement works when we try to quickly and accurately point to or tap on a target

Fitts' Law is usually written as:

Time	Distance
$T = a + b \log_2 (2 D/W)$	
Coefficients	Width

Index of Difficulty or ID

- **T** = movement time
 - **D** = distance to the target
 - **W** = width of the target (how big it is)
 - **a, b** = empirically determined constants
-
- Original equation: $ID = \log_2(2A/W)$
 - Shannon-style reformulation:
 $ID = \log_2(A/W + 1)$, or $ID = \log_2(A/W + 0.5)$ (when $ID < 3$ bits)

Why a Logarithm?

1. Speed-accuracy tradeoff

When you try to move quickly to a target, it's harder to be accurate. If the target is farther away (**D** increases) or smaller (**W** decreases), it gets harder. But the difficulty doesn't increase linearly; it increases more slowly as you keep making things harder

2. Information theory

Paul Fitts was inspired by Shannon's information theory (like how much information can be sent through a noisy channel). He saw pointing as a process of "transmitting information" from brain to muscles. The logarithm comes from the way information capacity is calculated in bits

Time	Distance
$T = a + b \log_2 (2 D/W)$	
Coefficients	Width

Why a Logarithm?

3. Empirical fit

Fitts plotted how long it took people to point at targets of various sizes and distances. Found that a logarithmic relationship fits the data better than linear

4. Doubling/halving effect

The logarithm means that doubling the distance or halving the width increases the difficulty by a fixed amount. For example:

- Double the distance (**D**), log term increases by 1
- Halve the width (**W**), log term also increases by 1

Making it twice as hard doesn't make users take twice as long, but just a little bit

Time	Distance
$T = a + b \log_2 (2 D/W)$	
Coefficients	Width

Why a Logarithm?

The logarithm in Fitts' Law reflects:

- The way difficulty increases with distance and decreases with target size
- The information-theoretic basis of the law
- And the actual observed behaviour of people moving to targets

In short: The logarithm captures the diminishing returns in movement time as tasks get harder, matching both theory and experiment

Time	Distance
$T = a + b \log_2 (2 D/W)$	
Coefficients	Width

Output: response

❑ Motor

Fitts' Law

Fitts' law states that the amount of time required for a person to move a pointer (e.g., mouse cursor) to a target area is a function of the distance to the target divided by the size of the target

However, there is a trade-off between speed and accuracy:

- Faster is sloppier (inaccurate)
- Slower is precise (accurate)

Whether speed of reaction leads to reduced accuracy depends on the task and the user

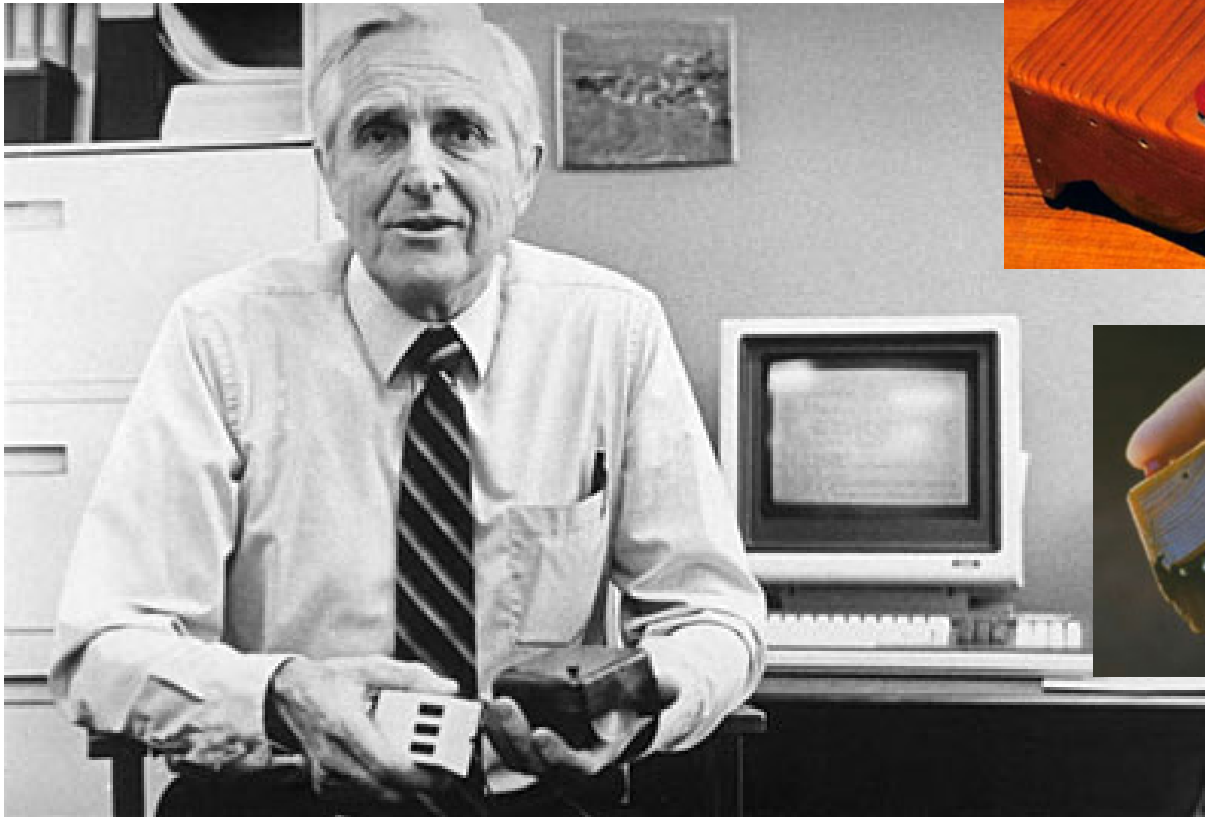
That is why you do training: fast + accurate



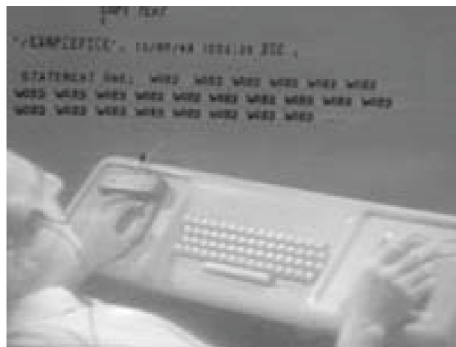
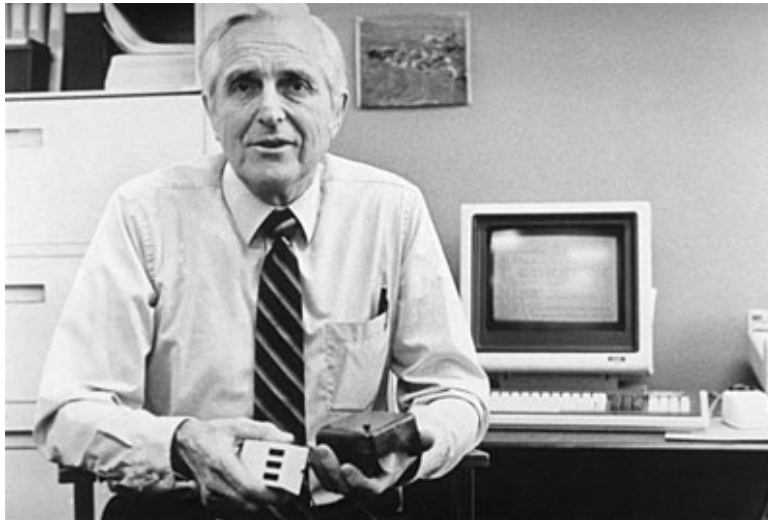
Fafa Yim, Logitech pro-team captain, *League of Legends*
fast and accurate



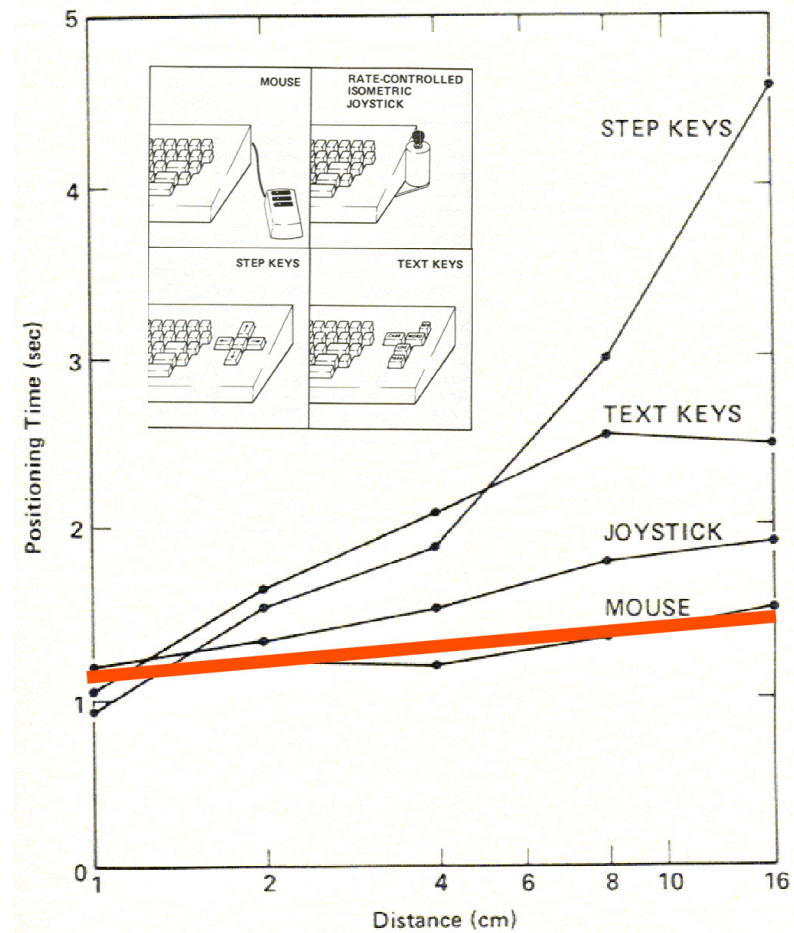
Douglas Engelbart (1964), inventor of the mouse



Douglas Engelbart (1964), inventor of the mouse and interaction researcher: Which is fastest?



Card, Stu. Lecture on Human Information Interaction. Stanford, 2007.



Output: response

❑ Motor

Fitts' Law

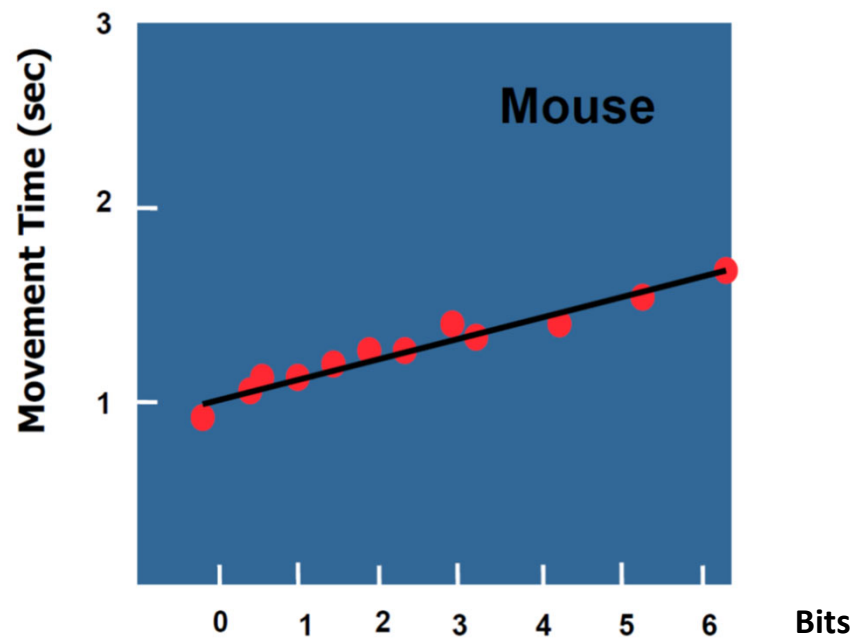
The channel is characteristic of the modality of the interaction

- a = y-intercept
- b = slope (secs/bit)
- T_{pos} = time taken (seconds)

Putting everything together:

- Time T_{pos} needed to complete the task is linear with respect to ID
- $T_{pos} = a + b \times ID$
- a and b are empirically determined through experimental data

Time	Distance
$T = a + b \log_2 (2 D/W)$	
Coefficients	Width



Output: response

❑ Motor

Fitts' Law

Evaluating a device

Index of Difficulty depends on the task,
not the device

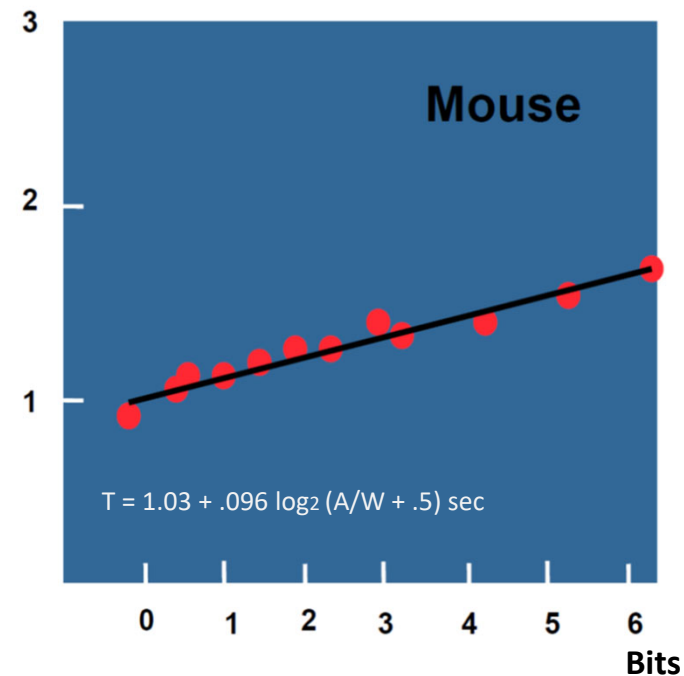
The regression constants a and b determine how
fast the task is done (or how fast the
device processes the task)

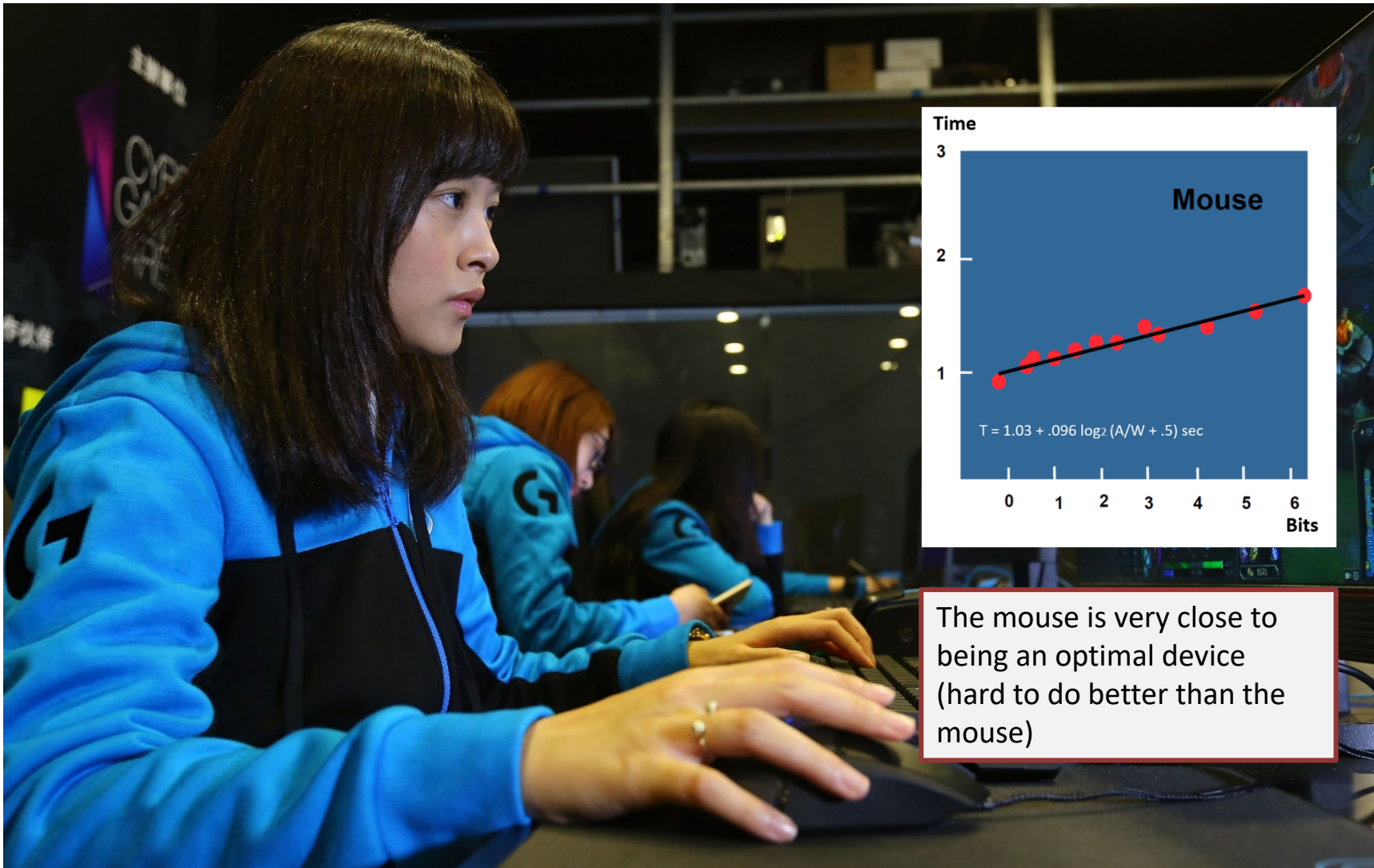
The *Index of Performance* IP (or *throughput* T_{pos})
defines how quickly the task can be finished

$IP = 1/b$

$$I_D = \log_2(\text{Dist/Size} + .5)$$

Time





Output: response

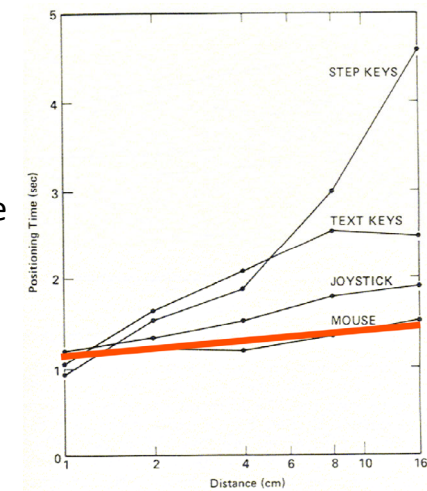
☐ Motor

Time scale of human behaviour

10^7 (months) 10^6 (weeks) 10^5 (days)	Social	Social Behavior
10^4 (hours) 10^3 10^2 (minutes)	Rational	Adaptive Behavior
10^1 10^0 (seconds) 10^{-1}	Cognitive	Immediate Behavior
10^{-2} 10^{-3} (milliseconds) 10^{-4}	Biological	

Attitude
formation

Performance



Output: response

❑ Motor

Time scale of human behaviour

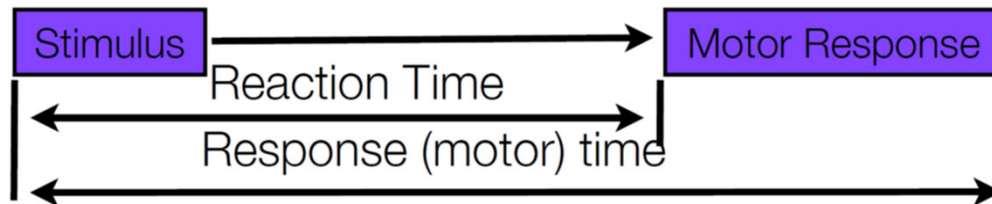
movement time
motor time

Time taken to respond to stimulus (speed): reaction time + response time

- Reaction time is dependent on stimulus type:

- Visual: ~ 200 ms
- Auditory: ~1500 ms
- Pain: ~ 700 ms

- Response time is dependent on physical characteristics of subject, like age, fitness level



Output: response

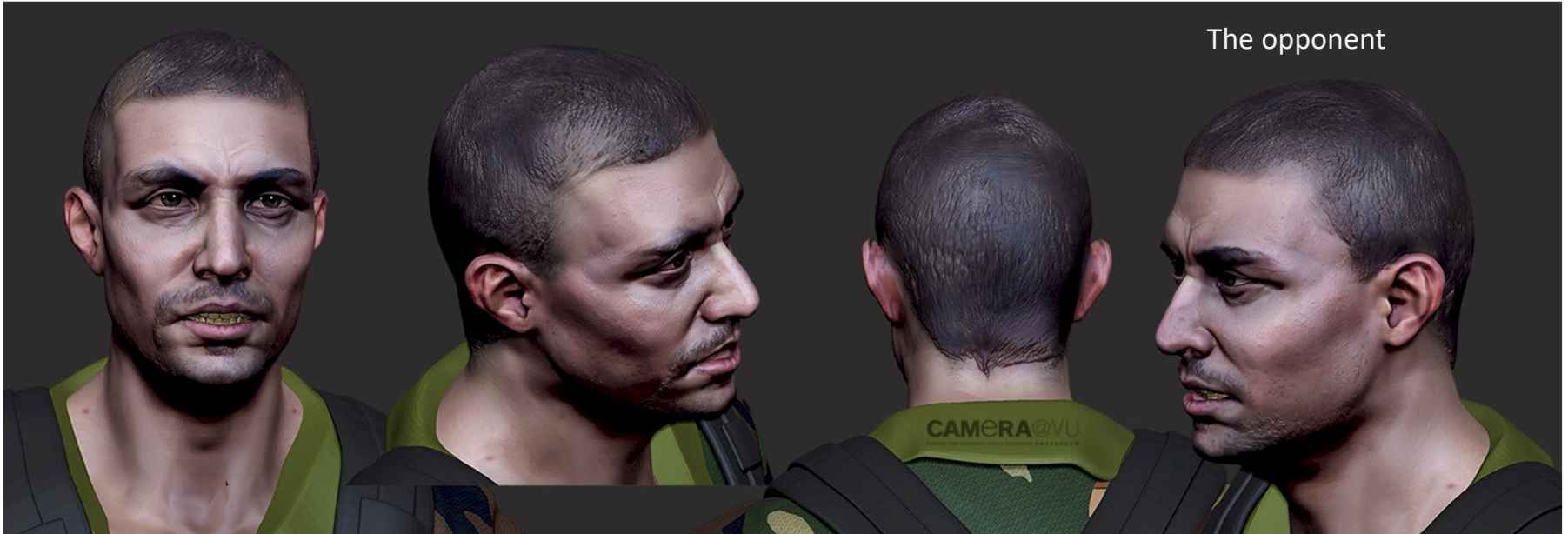
- ❑ Motor output: reaction times

Research into gaming

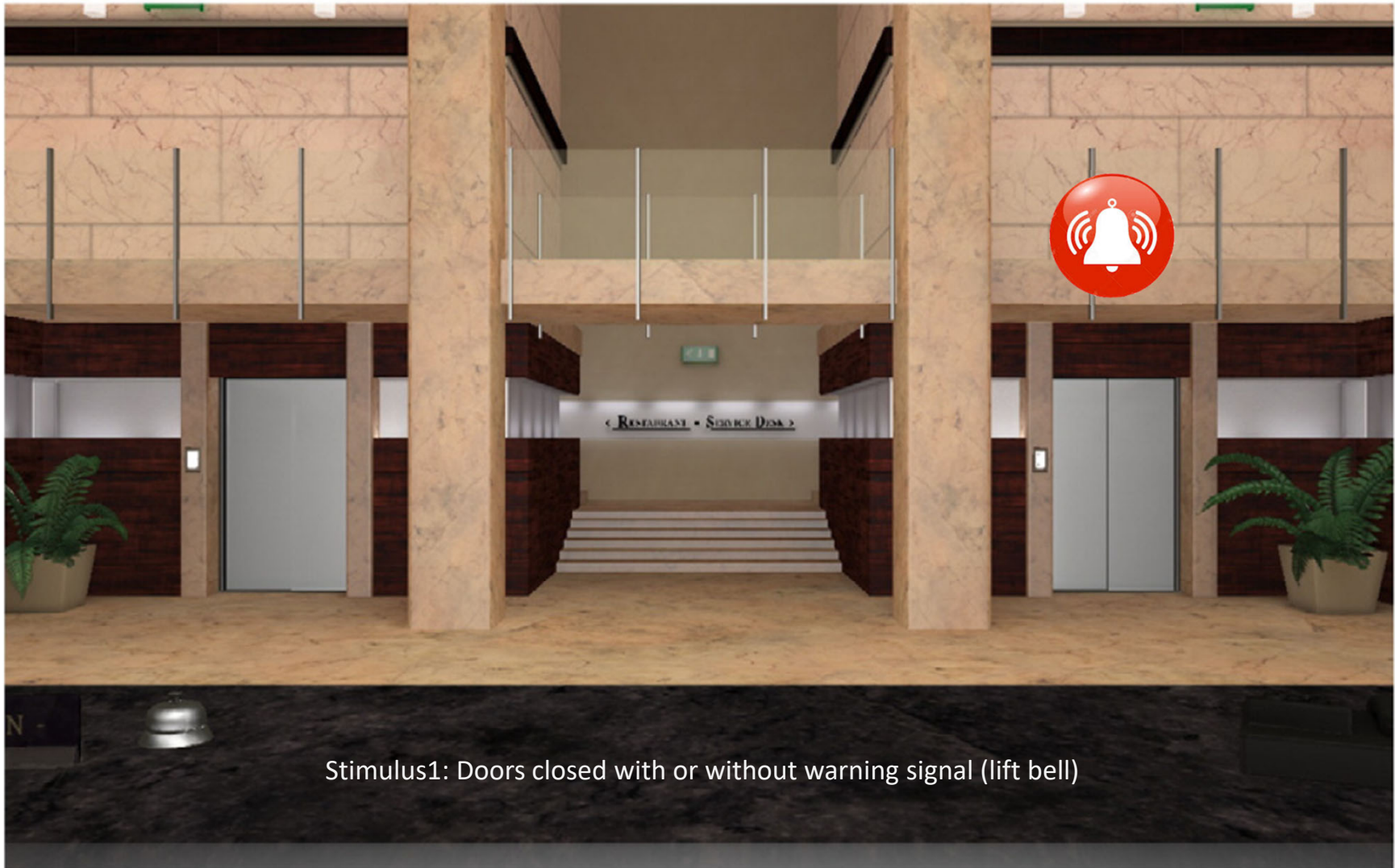
- Life-size large-screen projection of a hotel lobby
- Two elevators from which an opponent may enter
- Elevator door opens with or without warning
- Opened door shows an opponent or not (empty elevator)
- Participant shoots an infrared rifle
- Reaction time of pulling the trigger
- Accuracy of the shot (infrared coordinates)



The opponent



Infrared air rifle (Crosman Airguns, Model 1077)





Mental chronometry:

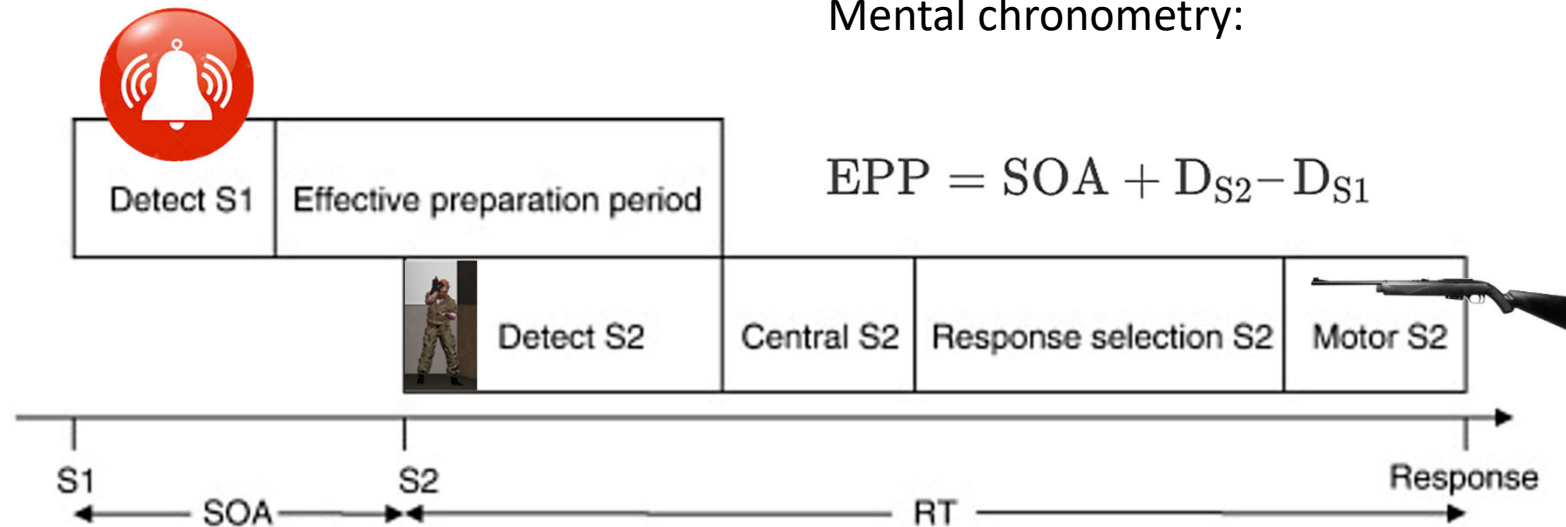


Fig. 1. A model of the time course of temporal preparation. It is assumed that temporal preparation occurs during the effective preparation period (EPP), which starts upon the detection of the warning signal (S1) and continues until the target stimulus (S2) has been detected. Preparation is assumed to facilitate central processing of S2. Note: SOA = stimulus onset asynchrony; RT = response time.

Use and Context

U1 Social Organization and Work



U3 Human-Machine Fit and Adaptation

U2 Application Areas

Meet ZHANG Wei, former student of COMP3423

Back in my village, I made a pill reminder app for my grandparents so the buttons should be big because they cannot see tiny characters. But on their Meizu M8, menus become pretty small. So I did some tests with them, figuring out several button sizes and menu options and how fast they could click – on average. Then I calculated Fitts for each layout and found the best click time as related to button size within the limitations of such small screens.

Red pocket money was good last Spring Festival.





Meet Vincy WONG, former student of COMP3423

At IT Global Support, I was in a team responsible for optimizing the efficiency of the business processes for our American and Asian divisions. But what you program is directly related to national, regional, and occupational work culture, I learned (the hard way). Efficiency for the USA division was geared towards process speed. In Singapore, it was about making no errors (I guess it's the guilt culture).

The US division allowed for shortcuts; Singapore followed the rulebook. We speeded up business processes for our US division by storing data on the Cloud and downloading open-source software our coders could adapt to new situations. For Singapore, we implemented new workflow automation with multiple control loops, allowing for fact checking and detecting incompleteness of forms and files.

Therefore, my recommendation for global support of 'efficiency,' the specs should be dynamic, able to move from fast to slow in a trade-off with more or less accuracy.



Human-generated Summary



Human Information Processing is different from computation because perception, memory, problem solving, and (motor) responses are less 'stable' but also more adaptive

All the visual, textual, sound, and touch input become an integrated whole and recede in memory, sometimes in a single cell (e.g., faces and places)

Memory is not fixed but changes over time (neurons change their connections). Memory is context-sensitive

Problem-solving skills may differ. For deterministic problems, you need reasoning (e.g., means-end analysis). For underdetermined problems, you need creative solutions

Human-generated Summary



When you design a device, you need to reckon with speed-accuracy trade-offs. That is, trained users can be highly efficient, combining high speed with great accuracy (cf. pro-team gamers, jet-fighter pilots)

It also means that untrained users (novices, first-time users, older adults) may not be as advanced as you

User testing of speed and accuracy tells you how your system or device does in terms of user performance. Don't forget to ask for appreciation as well (like-dislike)

Fitts' law as derived from Shannon's Theory of Information Systems is your means to determine user performance of motor output

Exam Sample Item

Category “Knowledge”

When can you consider a problem to be ‘well-defined’?

- A When goal decomposition allows to break down a complex problem into smaller ones by determining sub goals
- B When structured decisions are made in a problem state that has multiple representations
- C With full specification of the provided data, actions, strategies, and goal states, which can be represented in a problem space
- D When at the structural level, the focus lies on changes in structure and processes such as work units and actions

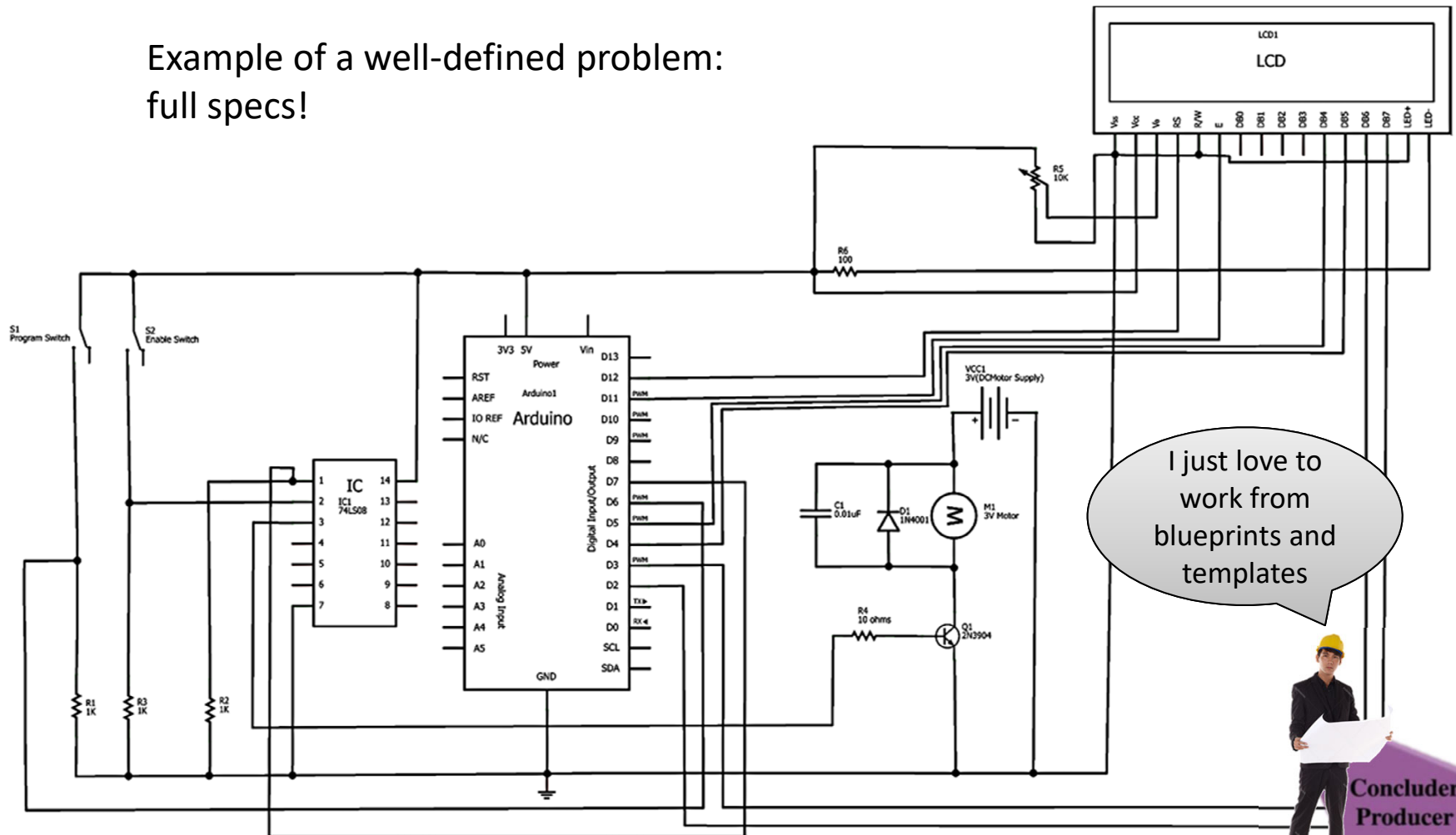
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Example of a well-defined problem:
full specs!



I just love to work from blueprints and templates



Concluder
Producer

In *Arduino Electronics Blueprints*, Don Wilcher explains how a programmable logic controller (PLC) is used to operate various electronic and electromechanical devices that are wired to I/O wiring modules (<https://hub.packtpub.com/programmable-dc-motor-controller-lcd/>)

References and recommended reading

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